

CHENYU (AUSTIN) ZHANG

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EDUCATION

Harvard University

Cambridge, MA

S.M. in Computational Science and Engineering Candidate, GSAS / SEAS

Sep 2026 – Dec 2027 (expected)

Relevant Coursework: Quantitative Methods for NLP, Advanced Scientific Computing, MLOps

Peking University

Beijing, China

B.S. in Statistics, School of Mathematical Sciences

Sep 2022 – Jun 2026

Relevant Coursework: Deep Learning and RL, Statistical Learning, Stochastic Processes, Mathematical Statistics

EXPERIENCE

Tencent

Beijing, China

Research Intern, Qingyun Program; Core Contributor, UI-Mate Technical Report

May 2026 – Aug 2026

- Owned task construction for **OSWorkerBench**, a benchmark of 100 long-horizon office workflows spanning 41 applications and 10 job families, released with **UI-Mate**, Tencent Hy Team's foundation GUI agent.
- Designed the capability-grounded task generation pipeline and made every task automatically scoreable by weighted checkpoints on final app state, then refined them through iterative human-in-the-loop verification using pilot-agent rollouts and synthetic test cases.
- Recorded human demonstrations for hard tasks, making this the first computer-use benchmark to supply multimodal demonstrations as one-shot guidance and to test procedural transfer from related but non-identical source tasks.

Carnegie Mellon University, Machine Learning Department

Pittsburgh, PA

Research Associate, advised by Prof. Katerina Fragkiadaki

Jul 2025 – Nov 2025

RobotArena: unlimited robot benchmarking via real-to-sim translation

ICLR 2026

- Built a real-to-sim pipeline converting teleoperation video into physics-consistent 3D environments via generative scene modeling and differentiable rendering.
- Designed automatic camera-robot calibration using 3D Gaussian Splatting, reaching sub-centimeter pose error with no manual supervision; applied it to BridgeV2, DROID, and RH20T obtaining more than 10k calibrated trajectories.

ParticleDiffuser: generative 3D object-particle dynamics

- Developed a diffusion model predicting 3D object-particle and gripper trajectories from point clouds, covering rigid and deformable objects.
- Built a goal-conditioned guided-diffusion controller with **3×lower MSE and 20×faster planning** than an MPC baseline; extended guidance across prediction chunks for long-horizon planning, and validated sim-to-real transfer.

Peking University

Beijing, China

Research Assistant, advised by Prof. Jiahuan Zhou

Nov 2023 – Jan 2025

SCAP: supportive clique-based attribute prompting for test-time adaptation

CVPR 2025

- Preserved vision-language model accuracy under deployment distribution shift with no labels and no retraining by adapting the model dynamically based on cliques of visually similar samples, achieving further improved performance on larger batches.

D2ART: dual-modal prompting for vision-language models

- Adapted CLIP at test time using entropy-guided prompt updates and EMA feature stabilization, improving top-1 accuracy by **2.3%** on ImageNet-A over prior test-time adaptation methods.

PROJECTS & SKILLS

Multi-Skill Dexterous Manipulation: RL to flow-matching, advised by Dr. He Wang

Jan 2026 - May 2026

- Built a three-stage learning pipeline for dexterous grasping: a flow-matching pose proposer conditioned on object point clouds, an RL pick-up policy rewarded against proposed grasps to prevent mode collapse, and a visuomotor student distilled from RL demonstrations.
- Injected noise during demonstration collection to collect recovery behaviors from the teacher, addressing behavioral-cloning distribution shift; reached **85%+ success** across Dexonomy with diverse grasp strategies.

Throughline: goal-tracking planner that turns notes and meetings into todos

Jun 2026 - Jul 2026

- Built and deployed a planning app tracking long/short-term goals: a pre-filter narrows candidates before the model adjudicates each proposed item as new, duplicate, or an in-place update against the live task list.
- Designed the *Profile* that represents the user's background, goals, and opinions, enabling the app to analyze notes and todos in the user's perspective to produce tailored advice. Implemented all app functions around the profile.

Languages: Python, MATLAB, R

Tools & Infrastructure: PyTorch, Git/GitHub, Linux, GPU cluster job scheduling, Claude Code, Fly.io, Neon

Interests: Tennis, Pickleball, Basketball, Piano, Chess, Football, Skiing